BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (MCQ's)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. $\int (ax + b)/(ax^2 + 2bx + c)\, dx =$ _____

(a) $2 \ln|ax^2+2bx+c| + c$
(b) $\ln|ax + b| + c$
(c) $\ln|ax^2+2bx+c| + c$
(d) $(1/2) \ln|ax^2+2bx+c| + c$

Q2. If $A = \int\sqrt{(1-\cos^2 x)}\, dx$, then A = _____

(a) 0
(b) 2
(c) -2
(d) None of these

Q3. The point of concurrent of altituds of a triangle is called _____

(a) Circum-centre
(b) In-centre
(c) Centroid
(d) Orthocentre

Q4. $ax^2 + 2hxy + by^2 = 0$ represent a pair of orthogonal lines If sum of the coefficients of x and y is:

(a) 1
(b) 0
(c) -1
(d) 2

Q5. If $90° < \theta < 180°$, then slope is:

(a) Undefined
(b) Positive
(c) Negative
(d) Same

Q6. lim (1 − 3x)^(2/x) = _____

(a) e^3
(b) e^4
(c) e^{-6}
(d) e^{-3}

Q7. A function f: A → B is said to be _____, if the range of f is B.

(a) Bijective
(b) Surjective
(c) Injective
(d) None of these

Q8. ∫ 1/(x ln x) dx = _____.

(a) ln|x| + c
(b) ln|lnx| + c
(c) lnx + c
(d) (1/x) lnx + c

Q9. The distance between two parallel lines 4x + 3y + 2 = 0 and 4x + 3y + 8 = 0 is:

(a) 13/2
(b) 59/2
(c) 6/5
(d) 2/5

Q10. If the slope of tangent is √3, then the angle is:

(a) 30°
(b) 60°
(c) 45°
(d) None of these

Q11. (1, 5) ∩ [3, 7]

(a) (1, 7]
(b) (3, 5)
(c) [3, 5)
(d) (3, 5]

Q12. If f(x) = √(x+2) then range of f is:

(a) (0, ∞)
(b) (−∞, +∞)
(c) (−2, +∞)
(d) [−2, 2]

Q13. If y = 5e^(3x−4), then dy/dx = _____.

(a) 15e^(3x−4)
(b) 5e^(3x+4)
(c) e^(3x−4)/3
(d) (3x − 4)e^(3x−4)

Q14. The vertex of the parabola y = 5(x−2)² + 7 is:

(a) (2, 0)
(b) (2, −7)
(c) (−2, 7)
(d) (2, 7)

Q15. The set S(c, r) = {p(x,y): | |cp| = r} contains centre c if:

(a) r = 0
(b) r < 0
(c) r > 0
(d) r = ∞

Q16. ∫ e^(3x)(3 sin x + cos x) dx = _____.

(a) e^(3x) sin x + c
(b) 3e^(3x) sin x + c
(c) e^(3x) cos x + c
(d) None fo these

Q17. If u, v, w are contermious edges of a parallelepiped, then its volume is:

(a) $|u \times v|$ (b) $[u\ v\ w]$

(c) $u \cdot v \cdot w$ (d) $u \times v \times w$

Q18. The centre of circle $3x^2 + 3y^2 + 6x + 12y - 9 = 0$ is:

(a) $(2, -4)$ (b) $(1, -2)$

(c) $(-2, 4)$ (d) $(-1, 2)$

Section-B

> Note: Solve any TEN of the following questions. Each question carries 05 marks.

Q2. Determine the limit of the sequence $(1 \cdot 2)/(3 \cdot 4)$, $(3 \cdot 4)/(5 \cdot 6)$, $(5 \cdot 6)/(7 \cdot 8)$, ...

Q3. Evaluate: lim (x^n − c^n)/(x^n − c^m)

Q4. Find the point which is equidistant from (2, 1), (5, 2) and (3, 4).

Q5. Find the equation of Hyperbola with centre at origin, focus at (6, 0) and accentricity is 3.

Q6. The point $(2, -5)$ is a vertex of a square one of whose side lies on the line $x - 2y - 7 = 0$.

Q7. The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other. Show that $8h^2 - 9ab = 0$.

Q8. Find the derivative of √(sin x) by first principle.

Q9. If y = (tan t)^ln x. Find dy/dx

Q10. Find the extreme value of the function f given by $f(x) = (x - 1)(x^2 - 2x)$, $\forall\ x \in R$.

Q11. Intergrate: $\int 25 \tan^2 5x\ dx$

Q12. Find the equation of the circle passes through the points (0, 0), (3, 0) and (0, −4)

Q13. Find the angle between the positive y-axis and the vector $a = i + 3j + 2k$

SECTION – C (DESCRIPTIVE ANSWER)

Q14. (a) Calculate approximate value of cos 61° (b) Intigrate: $\int (2y\ dy)/((y^2+1)(y^2+3))$

Q15. (a) Find the equation of the perpendicular bisector of the segment joining (3, 4) and (11, 14). (b) Find the vertex, focus and equation of directrix of parabola $y^2 - 6y + 8x - 23 = 0$

Q16. (a) Find the area of the triangle formed by the lines $y - x = 0$; $y + x = 0$; $x - b = 0$ (b) Find the equation of the tangents to the circle $x^2 + y^2 = 10$ at the point whose abscissa is 3.